Econ 2 - Lecture 6 - 1/26/26

Today: CPI & Unemployment (Chapters 3.2 & 3.3)

Wednesday: Unemployment + Review

Lecture Quiz #4 Released, Due on Monday, Feb. 2nd

Monday February 2nd: Midterm Exam → 30 MC questions

Lots of Resources! Write your own questions

Weekly Review Session on Fridays!

Last Class: Stable Prices

Policymakers target 2% inflation rate

→ Avoid deflation

→ Low, predictable inflation rate can help credit & labor markets run more efficiently

GDP Deflator: Measure of prices / inflation using components of GDP

Do consumers care about everything in GDP?

Consumer Price Index (CPI)

Only consider goods / services relevant to households

Step 1: What goods / services should we put in basket?

→ Food, Housing, Gas, Coffee, Phones, Eggs, education, etc.

Equivalent consumption (C) of GDP

→ Used goods: Cars, homes (?), computer hardware

Not counted in GDP → households care about prices

→ Imported Goods: Wine from France, sold in CA
Price matters to CA → include imports (M)

→ Exported Goods: Wine from CA, sold in France?
Not relevant to CA → exclude exports (X)

→ Gov't Purchases (G) & Investments (I)

→ Stop Sign Price?
Do not add to basket

→ Large Farm Tractor Price?
Do not add to basket

Step 2: Assign a weight to each good/service in basket → Represent the importance to average household budget

Lasik? In 2000, Lasik Price = \$10,000 / eye
In 2019, Lasik Price = \$2,000 / eye

↓

0.33%

80% Price Drop

Food = 14%, Energy = 6%, Gas = 3%
Housing = 35%

P × Q & sum

Step 3: Calculate weighted basket cost

Good	Base Year Weight / Q	Base Year P	Weighted Cost
Housing	30	100	$Q^H_{BY} \times P^H_{BY} = 30 \times 100 = 3000$
Food	50	10	$Q^F_{BY} \times P^F_{BY} = 50 \times 10 = 500$
Gas	20	4	$Q^G_{BY} \times P^G_{BY} = 20 \times 4 = 80$

Basket Cost in Base Year (2016) = 3000 + 500 + 80 = \$3580

In 2019?

BY = 2016

Good	Q_{BY}	P_{BY}	P_{2019}
Housing	30	100	120
Food	50	10	20
Gas	20	4	2

Weighted Basket Cost in 2019

$$= Q^H_{BY} \times P^H_{CY=19} + Q^F_{BY} \times P^F_{CY=19} + Q^G_{BY} \times P^G_{CY=19}$$

$$= 30 \times 120 + 50 \times 20 + 20 \times 2 = \$4640$$

CPI: Ratio of Basket Costs, relative to base year

→ Normalize the average price in base year to be 100! $CPI_{BY} = 100$

$$CPI_{CY} = \frac{\text{Weighted Basket Cost}_{CY}}{\text{Weighted Basket Cost}_{BY}} \times 100$$

$$= \frac{Q_{BY} \times P_{CY}}{Q_{BY} \times P_{BY}} \times 100$$

$$= \frac{4{,}640}{3{,}580} \times 100 = 129.60$$

$$\text{Inflation Rate}_{Yr\,A,\,Yr\,B} = \frac{CPI_{Yr\,B} - CPI_{Yr\,A}}{CPI_{Yr\,A}} \times 100$$

$CPI_{2020} = 258.1$, $CPI_{2025} = 326.03$

$$\text{Inf. Rate}_{2020,2025} = \frac{CPI_{25} - CPI_{20}}{CPI_{20}} \times 100$$

$$\text{Inf. Rate}_{20,25} = \frac{326.03 - 258.1}{258.1} \times 100$$

$= 26.3\%$ since 2020!

Base Year = 1982 - 1984

→ guessing!

Substitution Bias:

Eggs → Prices have increased 100% in 3 years

→ Cereal substitute

weight of cereal is too low

weight of eggs is too high

⇒ overestimate of CPI

⇒ 2% target is OK

→ minimize the possibility of a deflationary spiral

Goal #3: Full Employment

What causes unemployment?

Categorize Reasons for Unemployment

1. Frictional Unemployment: time lag in hiring process

To obtain a job: $\underbrace{\text{apply, interview, background checks, etc.}}_{\text{frictions}}$

New Graduates

Coral Tree worker → graduate in June, leave job.

New job in September → frictions

Not indicative of economic health

2. Seasonal Unemployment

Predictive annual patterns in employment

Not representative of economy

3. Structural Unemployment

Mismatch between skills of workers and employer needs

AI / technology: lead to long term job loss

⇒ Does not indicate a poor economy